

Home Lab

Windows Server, Active Directory, Windows 11 & Linux Integration

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 **Environment: VMware Workstation**

 **Domain: BIKSEC.com**

 **This document presents the design and implementation of a home lab environment simulating a small internal network using Windows Server, Windows 11, and Linux systems.**

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Project Overview

This project is a home lab setup built using VMware Workstation with multiple virtual machines running Windows Server 2022, Windows 11, and Kali Linux. The lab includes an Active Directory domain environment with DNS configured on the domain controller.

The environment consists of a domain controller (BIKSEC-DC01), an internal web server (BIKSEC-WEB01), a Windows 11 workstation (BIKSEC-WIN11), and a Linux system (BIKSEC-Linux). The systems are connected through a VMware NAT network and configured with static and DHCP-based IP addressing.

The lab includes domain creation (BIKSEC.com), domain joining of Windows systems, Linux network configuration using CLI tools, and basic connectivity testing between machines. Firewall configuration was adjusted to allow ICMP traffic for successful network testing.

Lab Components

System	Role
BIKSEC-DC01	Server (Domain Controller + DNS)
BIKSEC-WEB01	Server (Internal Web Server)
BIKSEC-WIN11	Workstation / Client Machine
BIKSEC-Linux	Linux System (Kali-based testing machine)

Tools Used

- VMware Workstation
- Windows Server 2022 (Desktop Experience)
- Windows 11
- Kali Linux
- PowerShell
- Linux CLI (nmcli)

Network Configuration

All machines are connected via VMware NAT network:

Subnet: 192.168.49.0/24

Gateway: 192.168.49.2

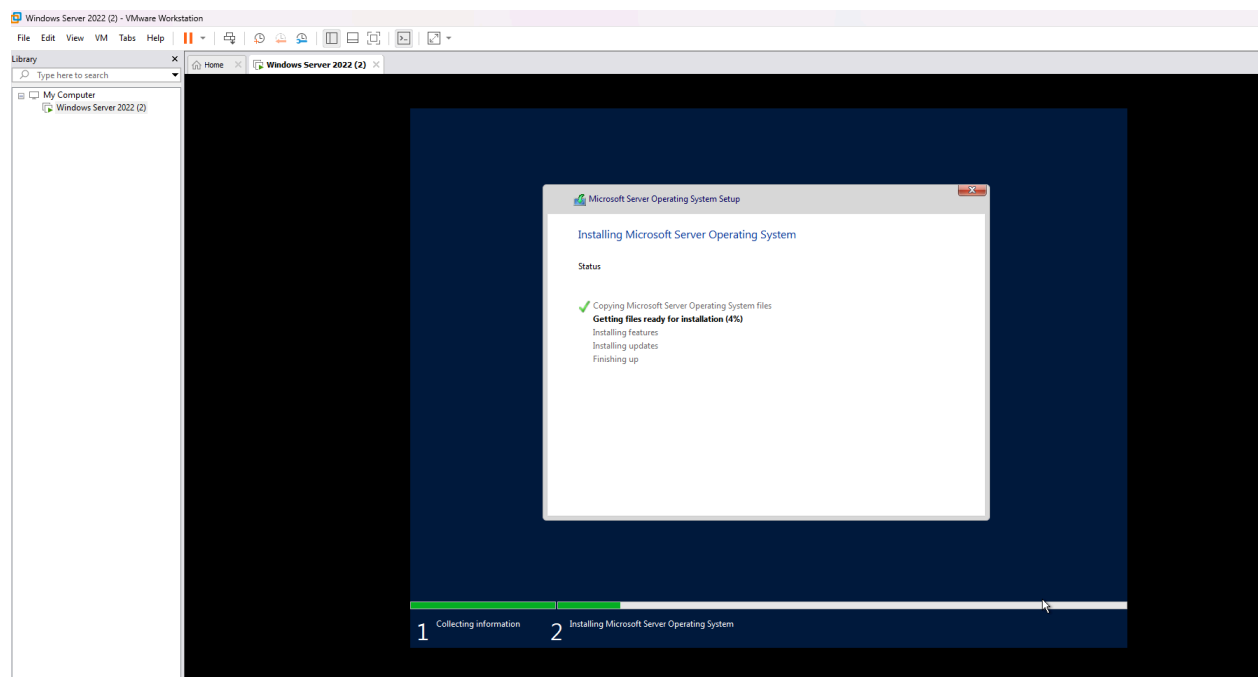
DNS: 192.168.49.10 (BIKSEC-DC01)

STEP 1 – Domain Controller Setup (BIKSEC-DC01)

Installation

- Installed Windows Server 2022 Standard (Desktop Experience)

 Screenshot: Windows Server installation



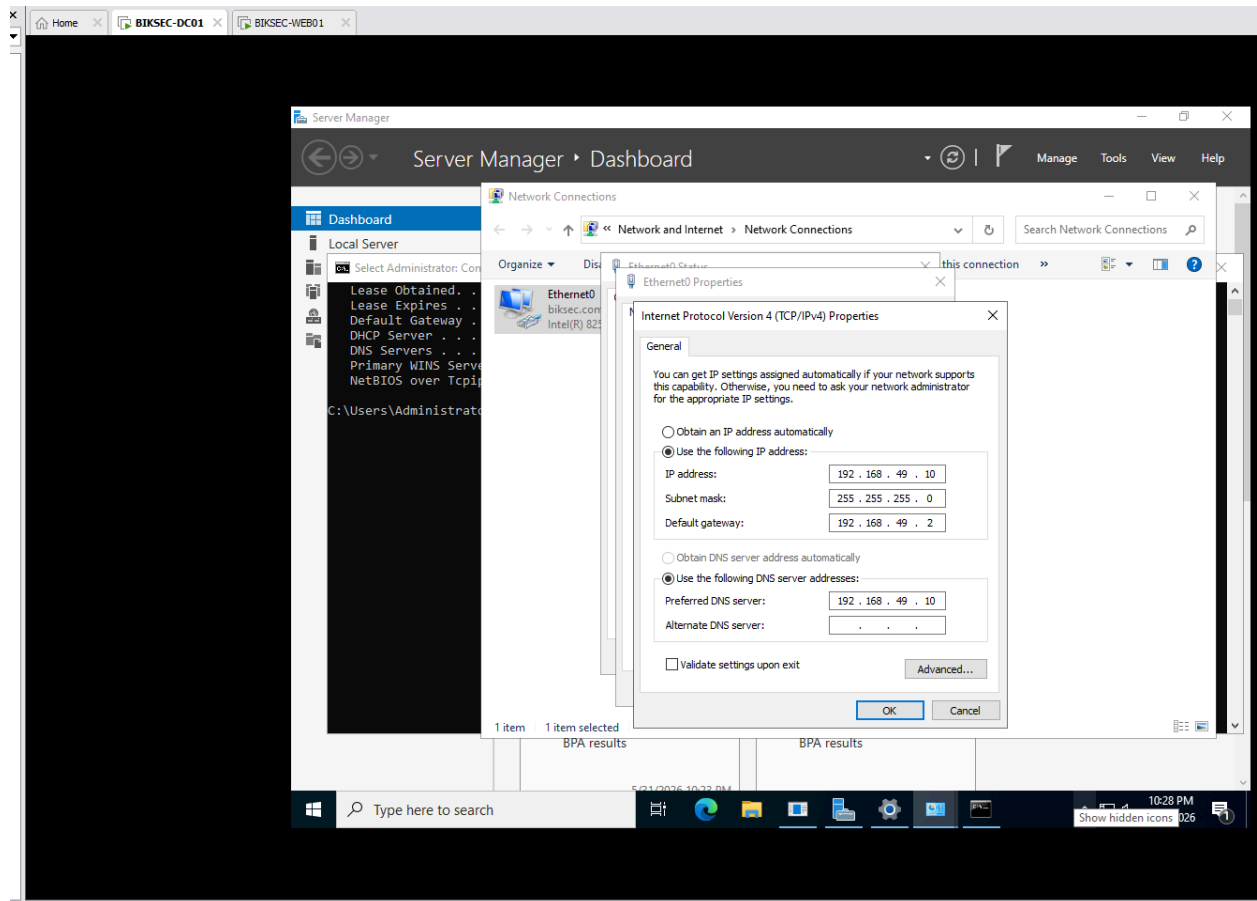
Static IP Configuration

IP Address: 192.168.49.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.49.2
DNS Server: 192.168.49.10

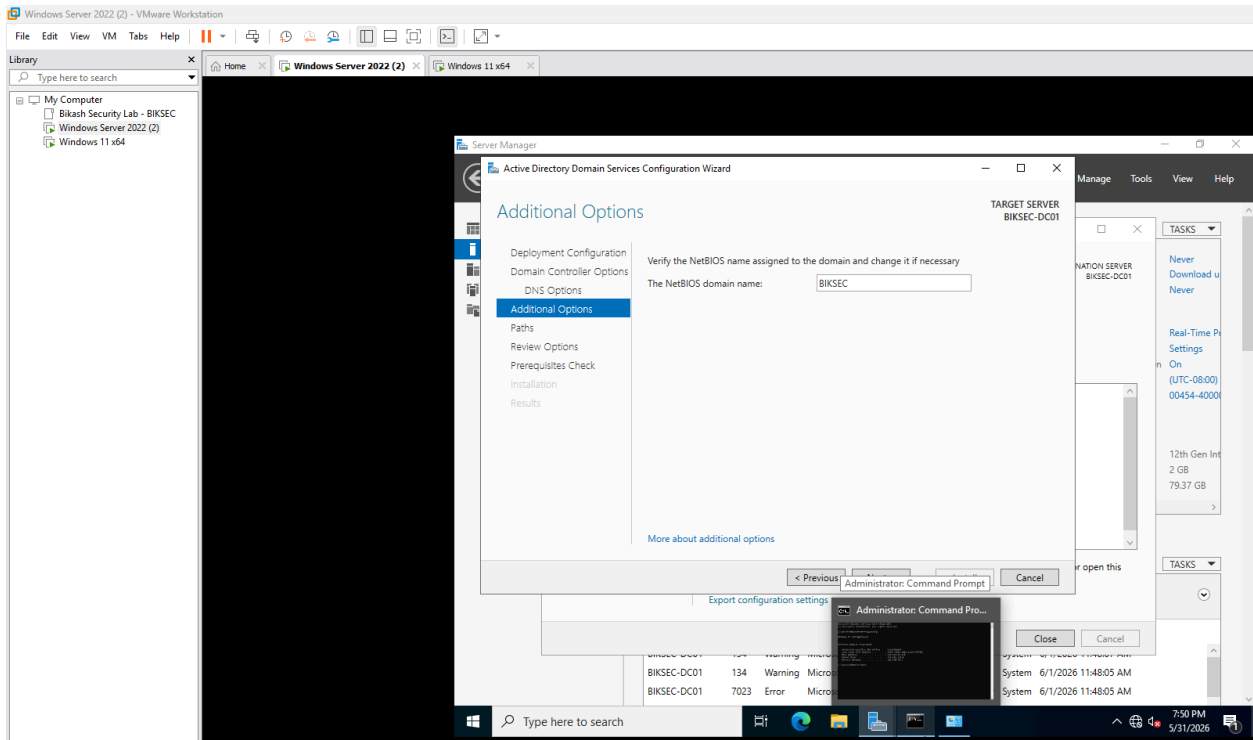
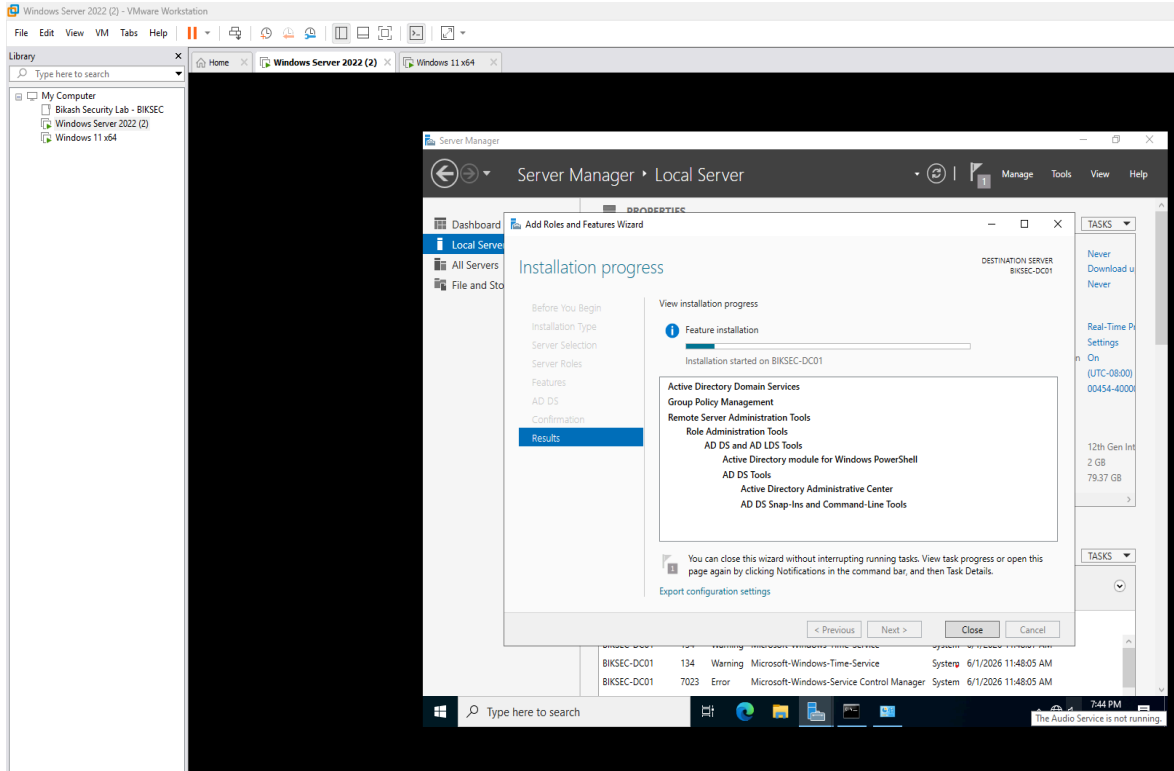
 Screenshot: Static IPv4 and DNS configuration



Active Directory Setup

- Installed Active Directory Domain Services (AD DS)
- Promoted server to Domain Controller
- Configured DNS role

 Screenshot: AD DS feature installation

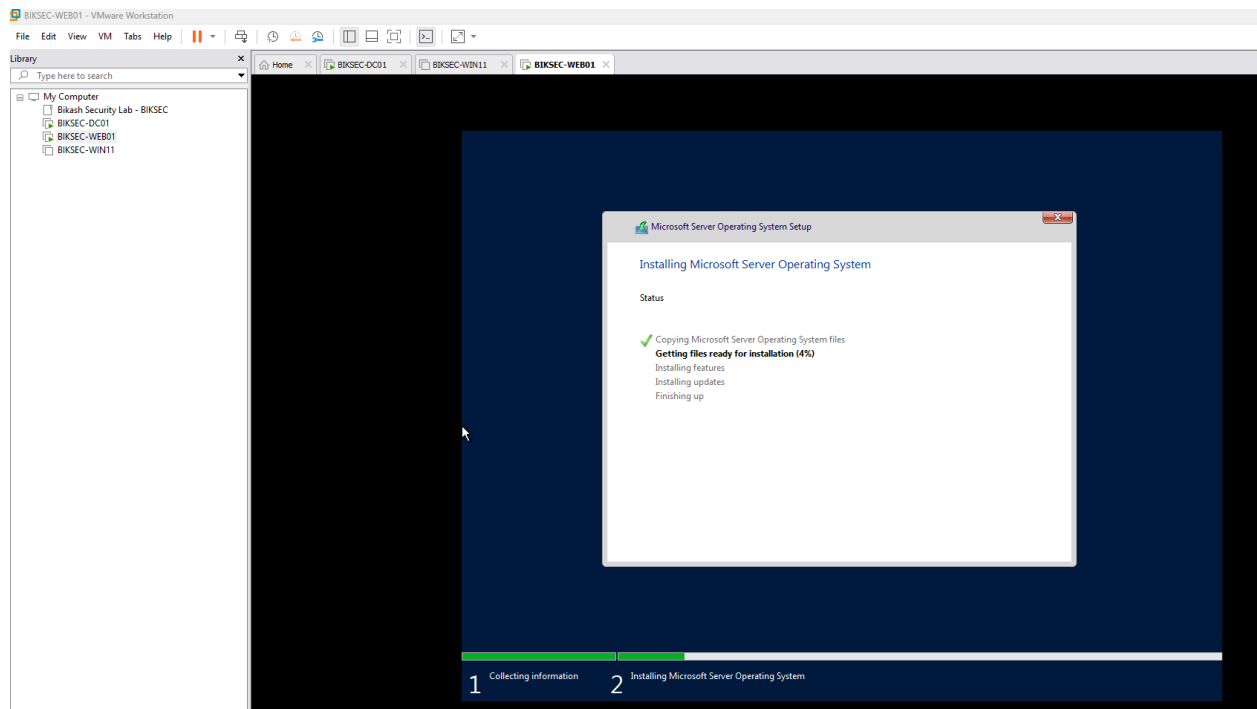


STEP 2 – Web Server Setup (BIKSEC-WEB01)

Installation

- Installed Windows Server 2022 Standard (Desktop Experience)

 Screenshot: Web Server installation



System Rename

- Renamed system to: **BIKSEC-WEB01**

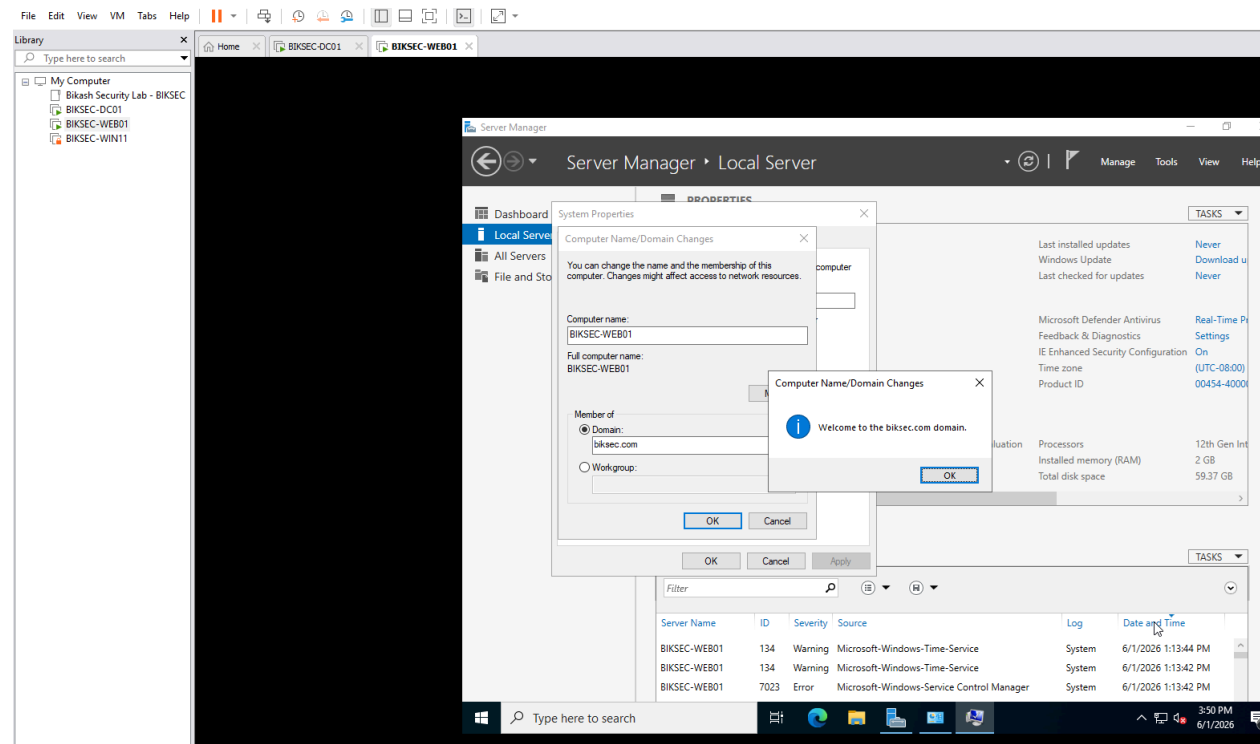
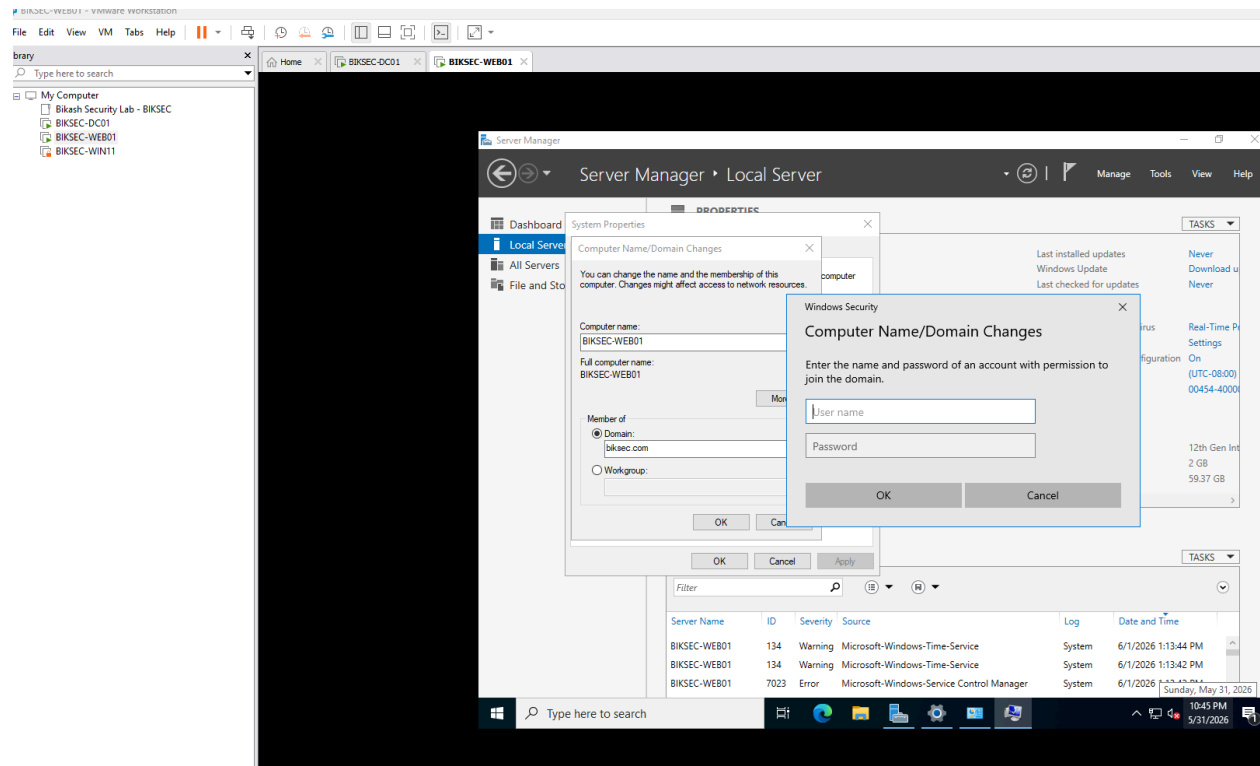
Static IP Configuration

IP Address: 192.168.49.20
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.49.2
DNS Server: 192.168.49.10

Domain Join

- Joined domain: **biksec.com**

 Screenshot: Domain join process

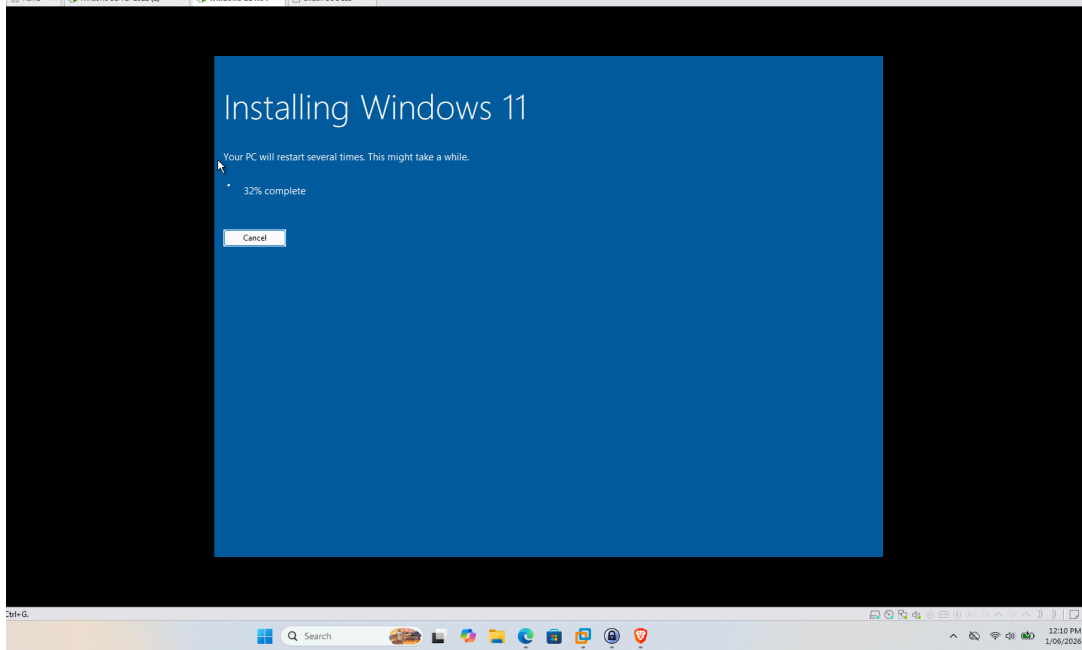


STEP 3 – Windows 11 Workstation (BIKSEC-WIN11)

Installation

- Installed Windows 11

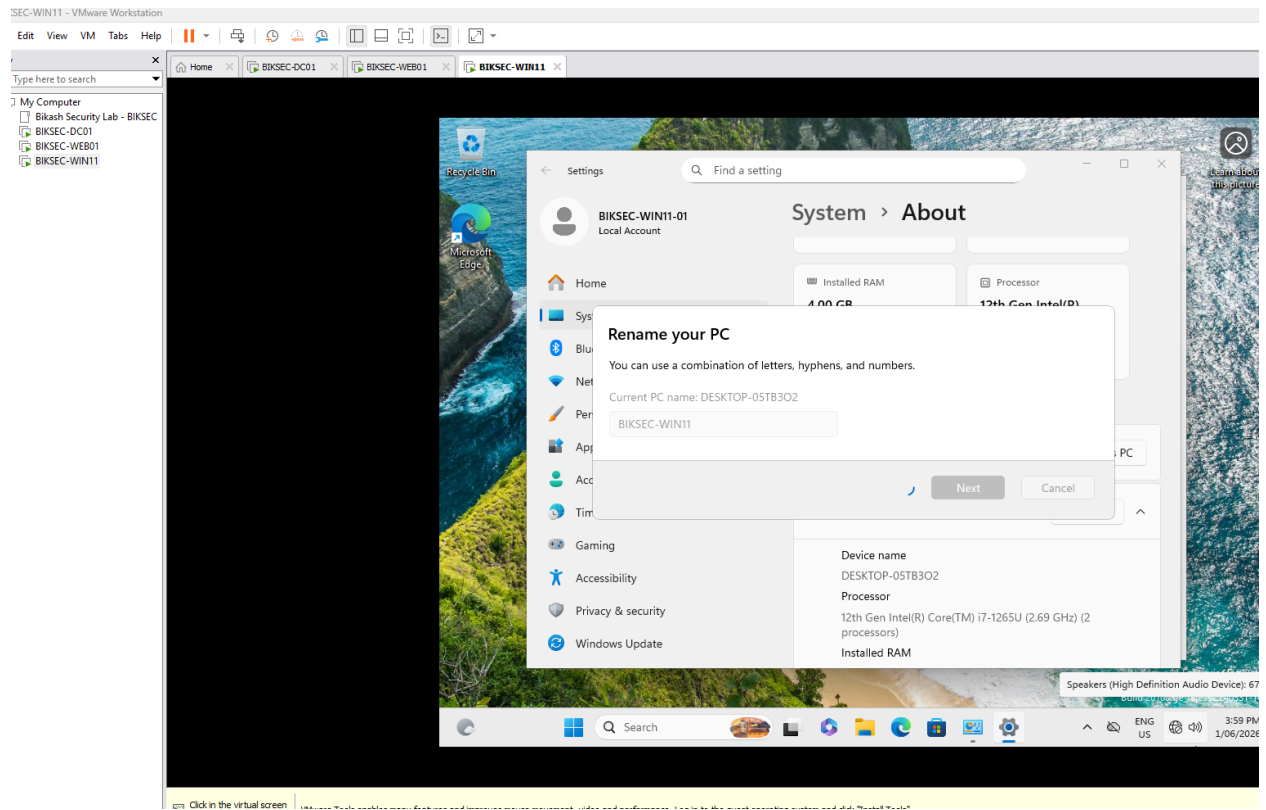
 Screenshot: Windows 11 installation



System Rename

- Renamed system to: **BIKSEC-WIN11**

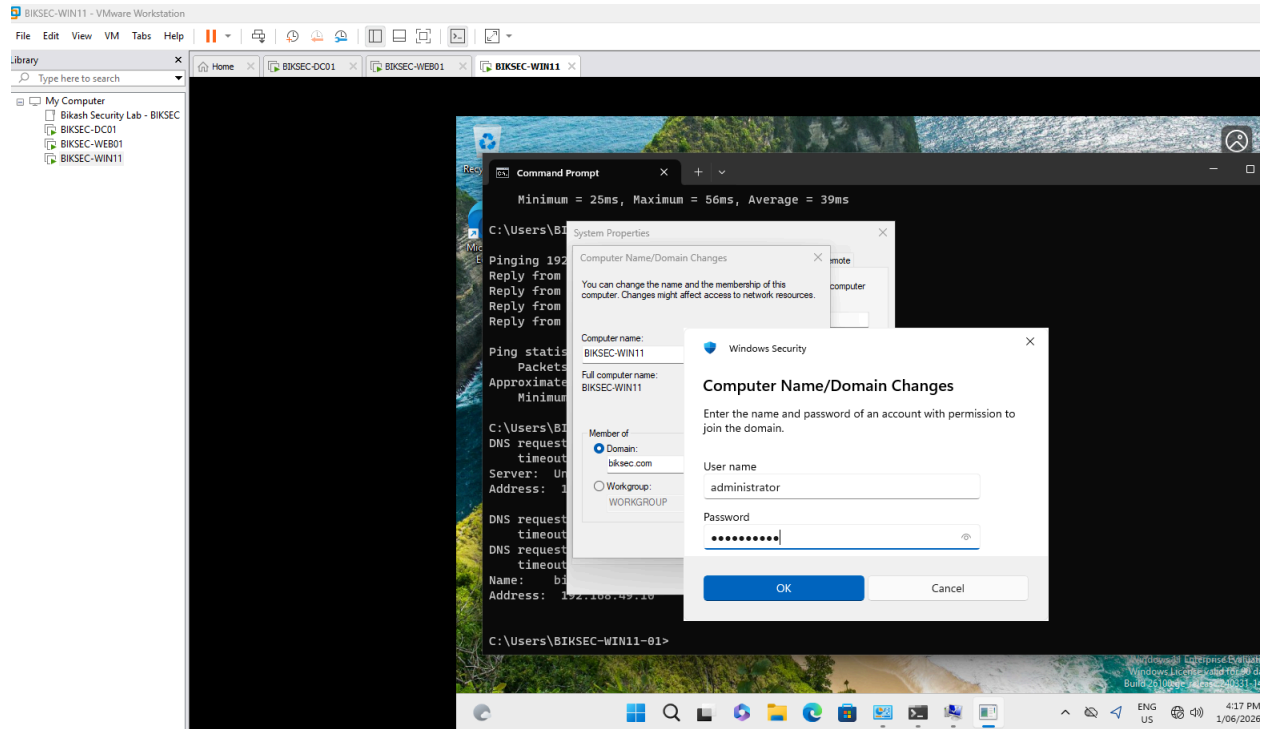
 Screenshot: System rename



Domain Join

- Joined domain: **biksec.com**

 Screenshot: Domain join process (Windows 11)

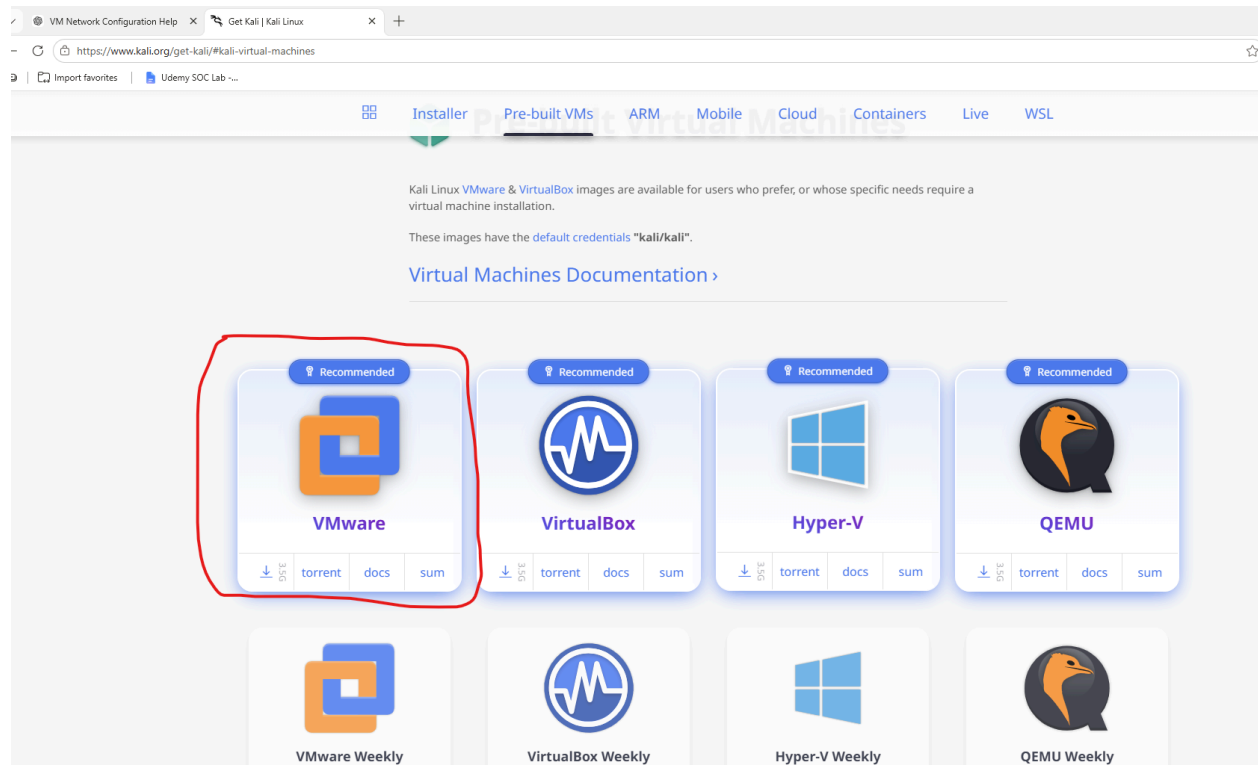


STEP 4 – Linux System (BIKSEC-Linux)

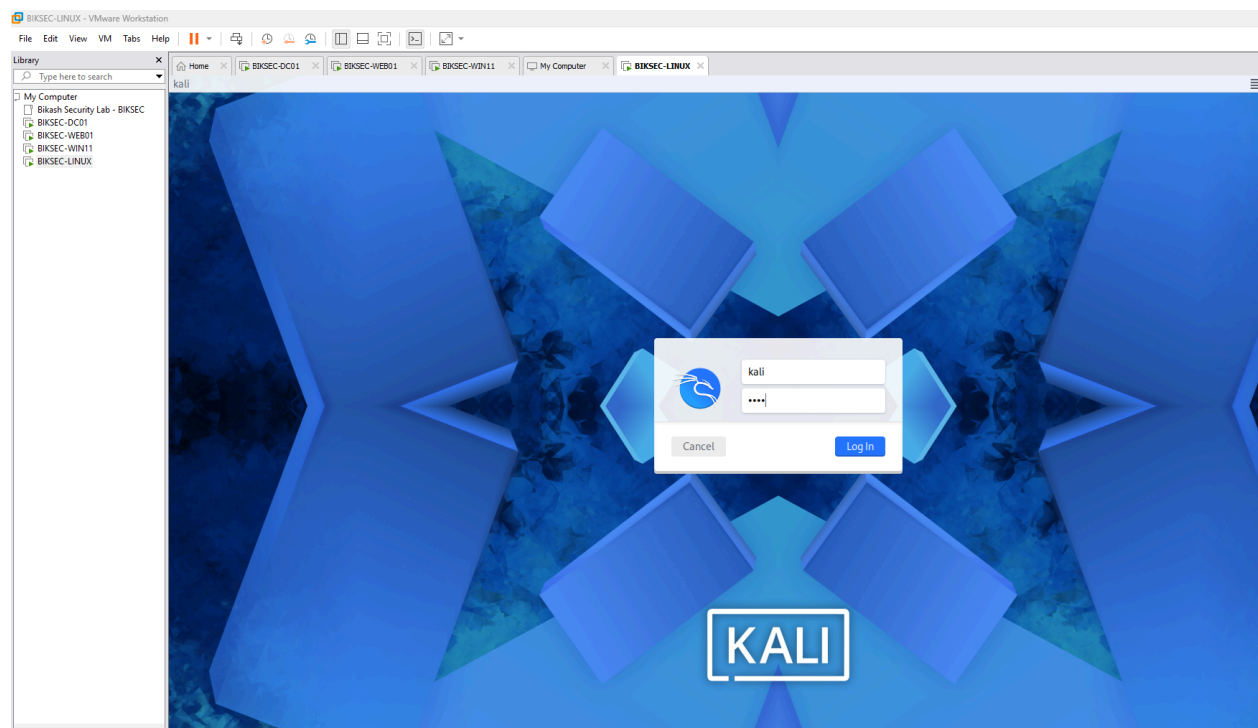
Installation

- Downloaded Kali Linux from official source
- Imported into VMware
- Default credentials:
 - Username: kali
 - Password: kali

 Screenshot: Kali Linux download page



 Screenshot: VM import and login



Static IP Configuration (CLI)

nmcli con show

nmcli con mod "Wired connection 1" ipv4.addresses 192.168.49.50/24

nmcli con mod "Wired connection 1" ipv4.gateway 192.168.49.2

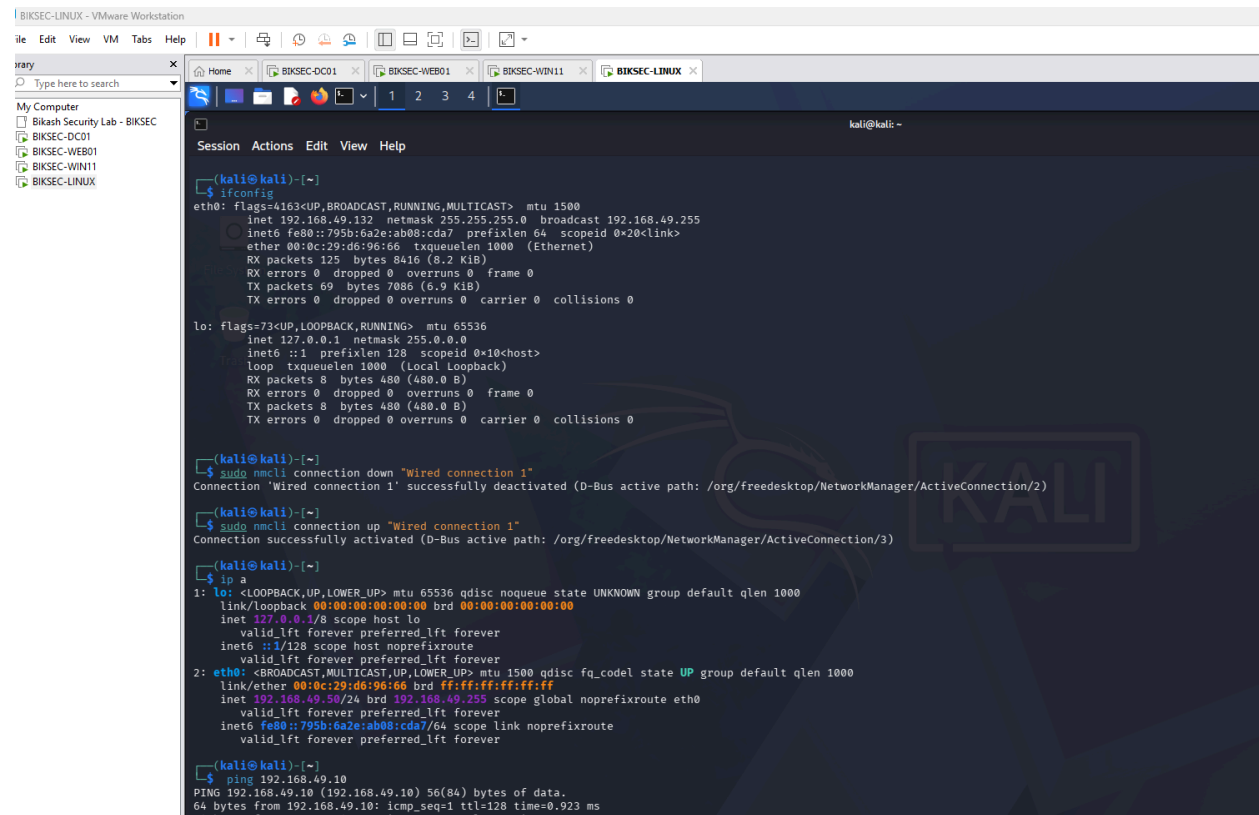
nmcli con mod "Wired connection 1" ipv4.dns 192.168.49.10

nmcli con mod "Wired connection 1" ipv4.method manual

nmcli con down "Wired connection 1"

nmcli con up "Wired connection 1"

 Screenshot: Linux IP configuration



```
(kali@kali)~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.49.132 netmask 255.255.255.0 broadcast 192.168.49.255
    inet6 fe80::795b:6a2e:ab08:cda7 prefixlen 64 scopeid 0<20<link>
    ether 00:0c:29:d6:96:66 txqueuelen 1000 (Ethernet)
    RX packets 125 bytes 8416 (8.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 69 bytes 7086 (6.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali@kali)~$ sudo nmcli connection down "Wired connection 1"
Connection 'Wired connection 1' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/2)

(kali@kali)~$ sudo nmcli connection up "Wired connection 1"
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/3)

(kali@kali)~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:d6:96:66 brd ff:ff:ff:ff:ff:ff
    inet 192.168.49.50/24 brd 192.168.49.255 scope global noprefixroute eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::795b:6a2e:ab08:cda7/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

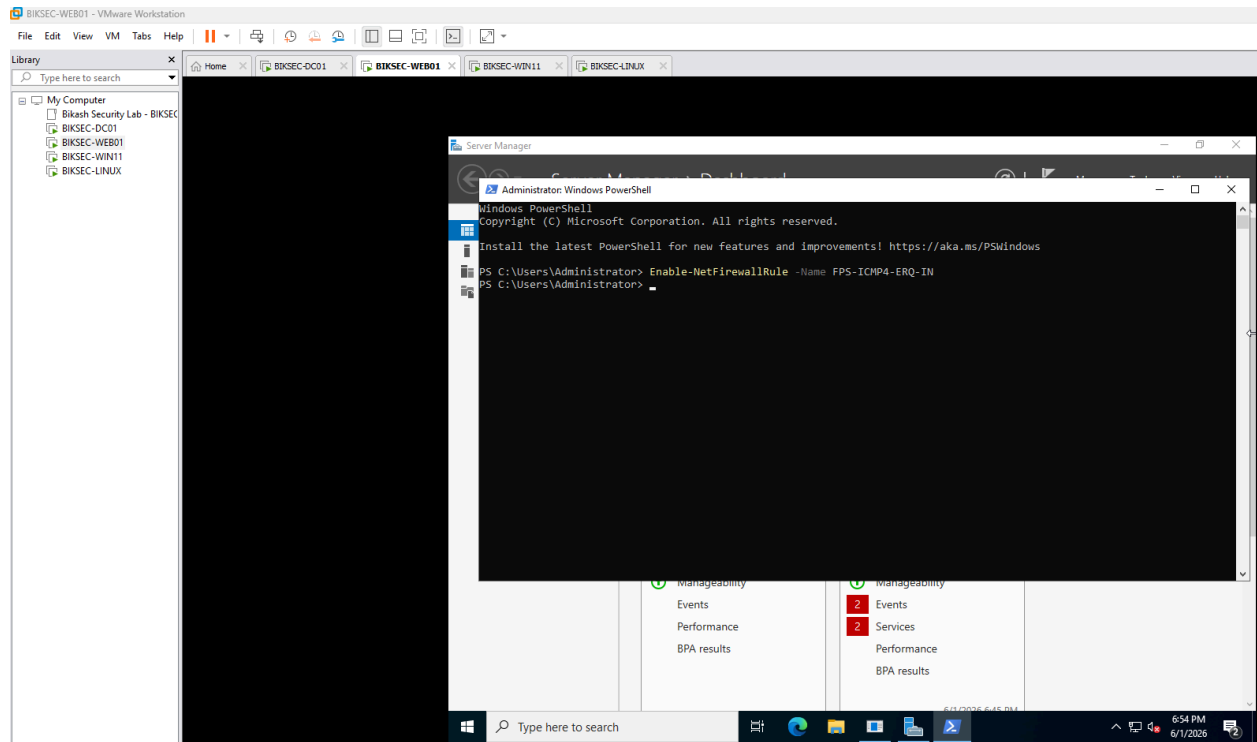
(kali@kali)~$ ping 192.168.49.10
PING 192.168.49.10 (192.168.49.10) 56(84) bytes of data:
64 bytes from 192.168.49.10: icmp_seq=1 ttl=128 time=0.923 ms
```

Connectivity Issue & Fix

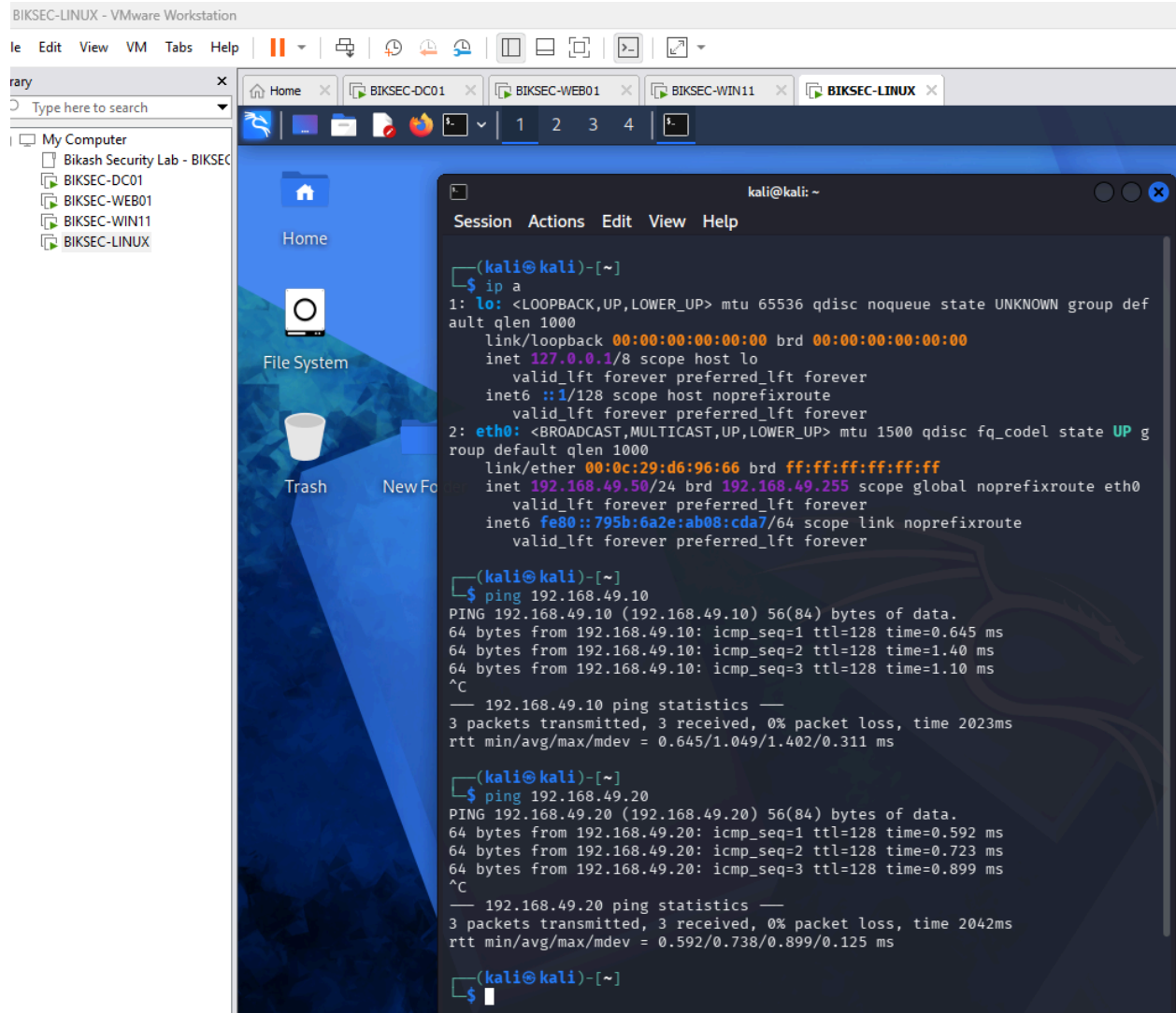
- Issue: Unable to ping BIKSEC-WEB01 from Linux
- Root Cause: Windows firewall blocking ICMP
- Fix applied:

Powershell cmd: Enable-NetFirewallRule -Name FPS-ICMP4-ERQ-IN

Screenshot: Firewall rule modification



Screenshot: Successful ping from Kali to Web Server



```
(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
    ault qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
    roup default qlen 1000
    link/ether 00:0c:29:d6:96:66 brd ff:ff:ff:ff:ff:ff
    inet 192.168.49.50/24 brd 192.168.49.255 scope global noprefixroute eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::795b:6a2e:ab08:cda7/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@kali)-[~]
$ ping 192.168.49.10
PING 192.168.49.10 (192.168.49.10) 56(84) bytes of data:
64 bytes from 192.168.49.10: icmp_seq=1 ttl=128 time=0.645 ms
64 bytes from 192.168.49.10: icmp_seq=2 ttl=128 time=1.40 ms
64 bytes from 192.168.49.10: icmp_seq=3 ttl=128 time=1.10 ms
^C
  192.168.49.10 ping statistics —
  3 packets transmitted, 3 received, 0% packet loss, time 2023ms
 rtt min/avg/max/mdev = 0.645/1.049/1.402/0.311 ms

(kali@kali)-[~]
$ ping 192.168.49.20
PING 192.168.49.20 (192.168.49.20) 56(84) bytes of data:
64 bytes from 192.168.49.20: icmp_seq=1 ttl=128 time=0.592 ms
64 bytes from 192.168.49.20: icmp_seq=2 ttl=128 time=0.723 ms
64 bytes from 192.168.49.20: icmp_seq=3 ttl=128 time=0.899 ms
^C
  192.168.49.20 ping statistics —
  3 packets transmitted, 3 received, 0% packet loss, time 2042ms
 rtt min/avg/max/mdev = 0.592/0.738/0.899/0.125 ms

(kali@kali)-[~]
$
```

Testing & Validation

- Verified inter-machine connectivity using ping
- Verified DNS resolution via DC01
- Confirmed domain join success for Windows and Web Server
- Validated Linux network access after firewall adjustment

Issues Encountered

1. ICMP Blocked on Windows Servers

- Cause: Windows Defender Firewall default policy
 - Fix: Enabled ICMP Echo Request rule via PowerShell
-

2. Static IP Configuration Errors

- Cause: Incorrect or missing gateway/DNS settings
 - Fix: Standardized network configuration across all machines
-

3. Linux Connectivity Issue

- Cause: Firewall blocking ping response on Windows
 - Fix: Enabled ICMP rule on target machine
-



Key Learning Outcomes

- Active Directory and domain-based network setup
 - Windows Server role configuration (AD DS + DNS)
 - Enterprise network segmentation using VMware NAT
 - Linux network configuration using nmcli
 - Firewall-based troubleshooting (ICMP control)
 - Cross-platform connectivity validation
-



Conclusion

This project demonstrates the successful setup of a home lab environment consisting of multiple systems, including Windows Server, Windows 11 workstation, and a Linux machine. The environment was configured with Active Directory Domain Services, DNS, and domain-joined systems to simulate a small internal network.

The lab helped reinforce practical understanding of network configuration, domain-based authentication, system administration, and cross-platform connectivity in a controlled virtual environment using VMware.
